

Mete Aksoy

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Education

Delft University of Technology, BSc in Computer Science and Engineering Delft, NL

- **Coursework:** Object Oriented Programming, Functional Programming, Data Mining, Big Data Processing, Machine Learning, Algorithms and Data Structures, Computer Graphics, Web and Database Technologies

Work Experience

Software Engineering Intern, Dutch Ministry of Defense, KIXS IT Innovation Unit Amersfoort, NL

- Developed AR-HUD systems on the **Magic Leap 2** platform. Implemented spatial anchoring for data overlays in dynamic and low-light conditions using **Unity**, integrated with **HLA-based** simulation software via **Pitch pRTI**.

Software Engineering Intern, PYM Automotive Istanbul, TR

- Supported the development and automation of software tools for automotive remanufacturing machinery, improving system efficiency and data tracking process by 8%.
- Collaborated with engineers to enhance machinery diagnostics and design workflows using **Python** and CAD-based tools.

Projects

LOB-Engine (Limit Order Book)

- Built a high-performance limit order book in **C++** for real-time trade matching using **std::map** for price levels and **std::list** for time-sequenced orders to maintain price-time priority.
- Optimized for efficiency with $O(1)$ order cancellations and $O(\log N)$ insertions, supporting both limit and market orders.

Housing Monitor Bot

- Built an automated housing listing monitor in **Python**, detecting new entries via API polling and sending real-time email alerts via **AWS EC2**.
- Integrated **Selenium** to automate applications to new listings after verifying compliance with `robots.txt`.

Integration of Trading Platform with Proprietary Options Application

- Developed a system in **Python** integrating **InteractiveBrokers** with a proprietary options trading application.
- Provided real-time critical stock option data including price, volatility, and **Greeks** via high-throughput data streams.
- Enhanced decision-making capabilities by ensuring timely and accurate market information for complex options trading strategies.

Splitser

- Implemented a group expense-settlement system using **Java Spring Boot** with a REST API and **JavaFX** client.

Data Mining Algorithms

- Applied dimensionality reduction (**SVD**), clustering (**K-Means++**, **DBSCAN**), and anomaly detection (**Isolation Forest**) using **NumPy** and **scikit-learn**.

Assembly x86-64 Hobby Projects

- Built a Brainf*ck interpreter, a custom `printf`, and bitmap-based message encryption in **x86-64 Assembly**.